[Logo: NCSC]

# PREDICTIVE IOT FRAMEWORK FOR EARLY POST-HARVEST SPOILAGE DETECTION

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## 1. INTRODUCTION: THE SILENT CRISIS

- **The Illusion of Safety:** Current warehouse inspections rely entirely on human vision. By the time decay is visible, it is already too late.
- **The Chain Reaction:** A single unnoticed rotting sack rapidly cross-contaminates surrounding healthy stock via moisture and fungal spread.
- **The Hidden Health Hazard:** Micro-spoilage that escapes the godown introduces severe biological toxins into the daily human food supply.
- **The Core Need:** We must shift defense mechanisms from a visual reaction to a chemical prevention.

**[Paste Image Here]**
**Visual depicting the scale of massive godown storage vs. the threat of invisible rot.**

## 2. HYPOTHESIS & CORE OBJECTIVES

*"If we monitor anaerobic off-gassing (Ethanol VOCs), we can predict spoilage up to 36 hours before physical decay."*

| | | | |
|---|---|---|---|
| 🎯 **Eradicate Waste** Intercept chain-reactions early | 👃 **Shift to Smelling** Chemical leading indicators | 💰 **Economic Scaling** Ultra-low cost (₹850/node) | 🛡️ **Secure Health** Block toxins from human supply |

## 3. EXECUTION METHODOLOGY

**[Icon/Photo: Field Survey]** → **[Icon/Photo: Hardware]** → **[Icon/Photo: Coding]** → **[Icon/Photo: Testing]**

**Phase 1: Field Research**

**Phase 2: MVP Engineering**

**Phase 3: AI Integration**

**Phase 4: Data Validation**

## 4. GROUND REALITY: FIELD VISIT GALLERY

[Photo: Kalupur APMC]

**Kalupur Grain Market:**
Surveyed visual inspection flaws.

[Photo: Naroda Godowns]

**Naroda Cold Storage:**
Analyzed poor ventilation impact.

[Photo: Asarwa Hub]

**Asarwa Sector:**
Observed baseline gas variables.

## 5. TECHNOLOGY & DATA ANALYSIS

- **Hardware (Edge Node):** Utilizes Arduino Uno & MQ-3 Sensor to detect anaerobic Ethanol/VOCs in real-time.
- **The "False Alarm" Problem:** Sudden spikes (like cleaning chemicals) must not trigger warehouse panics.
- **AI Solution (Random Forest):** Python algorithm tracks the *Moving Average*. It ignores sudden noise and only flags sustained biological decay trends.

**[Paste Technical Data Chart or App Screenshot Here]**
**Showcasing the AI Moving Average vs Raw Spikes**

## 6. THE LIVE WORKING MODEL

**[Paste the Largest, High-Quality Photo of your Physical Working Model Here]**

Clearly showing the MQ-3 Sensor, Enclosure, and LED/Buzzer wiring.

## 7. RESULTS & THE 36-HOUR ADVANTAGE

- The system successfully predicted spoilage in the testing enclosure without false alarms.
- Grants facility managers a **1.5-day intervention window** to isolate stock.
- Secures 100 sacks of inventory per node, ensuring an unmatched Return on Investment.

## 8. SDG IMPACT

- **SDG 2:** Zero Hunger (Preventing waste)
- **SDG 3:** Good Health (Stopping toxins)
- **SDG 12:** Responsible Consumption